

10.6.3 Learning Outcome 3

Use skills necessary to collaborate within a team on a software development project

- Collaborated effectively by sharing responsibilities and coordinating development tasks.
- Used Git for version control to manage code changes and maintain consistency.
- Communicated regularly to discuss progress, resolve technical challenges, and ensure alignment.
- Participated in joint testing and debugging to improve system performance and reliability.
- Engaged with supervisors and peers to gather feedback and refine the project.

10.6.4 Learning Outcome 4

Analyzed user-centered design, cognitive aspects, research methods, modelling, and prototyping with reflection on ethical, legal, professional, security, and social issues

- Applied user-centered design principles to create a simple and intuitive interface suitable for healthcare users.
- Considered cognitive aspects such as reducing complexity, improving clarity of outputs, and enhancing usability.
- Conducted basic research and reviewed existing systems to guide design and development decisions.
- Developed and refined prototypes iteratively to improve system usability and functionality.
- Implemented modelling techniques for CT-based tumor detection and MRI-based classification.
- Reflected on ethical and legal issues such as patient data privacy, responsible AI usage, and bias in predictions.

10.7. Problems and challenges faced

Data Preparation and Cleaning – Managing incomplete data records, inconsistencies, and variations in image quality were several challenges when preparing the data for the training. Serious whack was required to clean, normalize, and standardize the data to validate the interoperability with machine learning models and increase overall forecast accuracy.

Model Integration and Dependency – Dependency issues come into place when there is a need to integrate a number of trained models, which will later be integrated into one system, as all of them were implemented using the same libraries, though there are many versions. This created a need for environmental management and modification to validate smooth integration.